Shibboleth: Probe-and-Replay Model-Response Watermarking for Diffusion Language Models

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Abstract

We propose Shibboleth, a model-response watermarking method for masked diffusion language models (DLMs). Unlike an autoregressive sampler, a masked DLM can be re-evaluated on its own output under an arbitrary remasking of the context. Because a practical DLM, unlike an ideal predictor, shifts the likelihood of an emitted token by an amount its own weights determine, a watermarking channel opens in the relation between a text and the model rather than in the tokens. Shibboleth evaluates a keyed family of context probes against each masked block, admits as carriers only the unresolved positions whose block-masked conditional retains mass on both response signs, and resamples at those carriers from the live conditional until the requested sign is committed. Unlike logit-bias methods it leaves the model's probabilities untouched and conditions only on which model-drawn proposal is accepted, and unlike order-steering methods it spends none of the decoding schedule. Because the admitting gate is invariant to swapping two probes, it stays blind to the sign the key will request, and carriers may therefore be discovered adaptively without disturbing the null. Detection re-masks the finished text, rebuilds the identical probes and carriers, and counts keyed sign matches, whose distribution on text written without the key is exactly binomial, and p-values are therefore exact at finite sample size and under carriers the method itself chose, which no threshold fitted to a corpus can promise. It reaches 0.80 TPR at 1% FPR on LLaDA-8B-Instruct — below the strongest token-based detector there, at the only perplexity ratio in that comparison reading below its own unwatermarked baseline — and 0.90 on Dream-7B-Instruct, where every token-local baseline weakens under transfer and the model-response channel is the strongest detector measured. Verification in exchange costs one rerun of the checkpoint, and evidence written against a prefix degrades under editing — collapsing at low redundancy, halving at high — leaving the method suited to a verifier that holds the checkpoint rather than to open screening.

1 Introduction

Language models now produce text that readers cannot reliably separate from human writing, which turns provenance into a security problem, since disinformation, phishing, and academic fraud all grow cheaper once authorship can no longer be established [28]. Institutions facing that problem reach first for a post-hoc detector, and academic-integrity screening is the clearest case, given that an essay's authorship has to be settled after it is submitted [23]. A detector of that kind is nevertheless fitted to one generation process, and its verdicts weaken as deployment moves to models that decode by a different mechanism, masked diffusion models among them. Watermarking is the answer most widely deployed against it, since it plants a keyed statistical signal during generation and lets whoever holds the key test a later document for it [21, 32]. A watermark is nevertheless worth only the

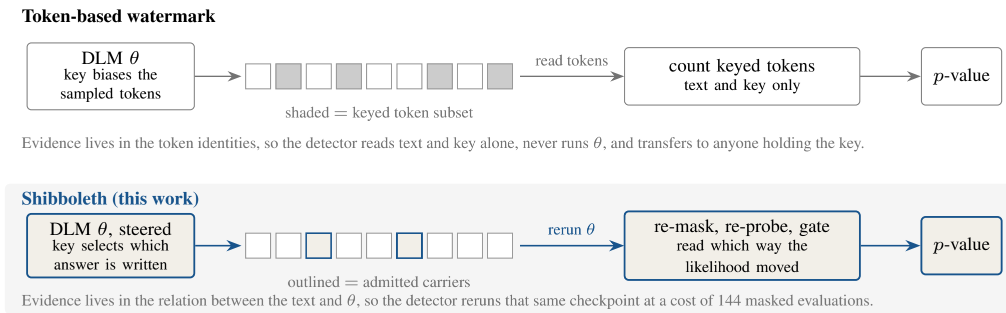

Figure 1: **Two places to keep a watermark.** A token-based watermark biases which tokens are sampled and recovers that bias by reading the text against the key. Shibboleth instead writes the answer the key requests to a keyed probe of the context, which makes the evidence a property of the text together with one checkpoint and obliges the detector to rerun it.

guarantee attached to its verdict, given that a false positive is an accusation against a human author, and a deployment therefore needs a false-positive rate it can state in advance rather than one fitted to a calibration corpus. Since adversaries also paraphrase output and learn a key's behaviour from detector queries [19, 23], the guarantees worth designing for are the ones that survive finite samples and adaptively chosen trials [10, 43].

Existing watermarks place their secret in the tokens, whether by biasing keyed vocabulary subsets [21], by coupling each draw to pseudorandomness [24], or, in recent diffusion language model (DLM) methods, by exploiting unresolved context and denoising order [17]. Detection then reduces to inspecting the text together with the key, and the generator is never consulted. Masked DLMs, by contrast, permit a different arrangement, since a finished text can be fed back to the model with selected context hidden and the likelihood of an emitted token then moves by an amount the checkpoint alone determines. Hiding a keyed subset of the context is in this sense a *context probe*, the direction in which the likelihood moves is the model's response to it, and provenance can be argued from the agreement between the two. Such a probe is worth applying only where the model has a genuine choice of answer, that is where several plausible tokens coexist at a masked position and react with opposite signs to two patterns.

Moving the secret out of the tokens and into that relation, however, costs two things a token-local detector obtains for nothing, since the verifier must apply exactly the probe the embedder applied, and the choice of carriers must be settled before the accepted answer is known; a gate with sight of the rewarded sign would otherwise inflate the false-positive rate above the level the detector reports. Section 3.2 states both as requirements, and existing DLM watermarks avoid them by keeping detection token-local, whether by removing prefix dependence [9], by averaging a red–green bias over unresolved context [14], or by steering which mask is resolved next [17], none of which asks the model anything about the text it produced.

Shibboleth instead meets each condition with one mechanism (Figure 1). It settles replayability by re-masking the current block together with its suffix before scoring the probe family, and the state it measures therefore follows from the final text alone. A *balanced gate* then admits as carriers only the unresolved positions whose block-masked conditional retains mass on both response signs, scoring them in a way that is invariant to swapping the two patterns and therefore blind to the sign the key will ask for. Writing the answer is the easier half, since Shibboleth draws at most R proposals from the unchanged live row at an admitted position and commits the first one carrying that sign, which amounts to rejection sampling and does change the output distribution, although every committed token remains one the model proposed. Verification afterwards re-applies the same probe, reapplies the gate, and returns an exact binomial tail. Unlike a prefix-based watermark, which can require strict left-to-right decoding, and unlike an order-guided one, which changes which mask is resolved next, Shibboleth acts only after the reference schedule has produced a row, and therefore leaves the block partition and the confidence order untouched.

- We show across 36 settings that this checkpoint's conditionals are too sharp for answers to be inserted after the fact, and the answer must therefore be written while the conditional is still live (Section 3.2).

- We propose Shibboleth, in which block-and-suffix re-masking makes the probe replayable from the finished text and orientation-blind gating keeps the accepted answer independent of how the carriers were discovered, which together keep the detector's p-values exact and free of calibration (Sections 3.3 to 3.5).
- We evaluate Shibboleth on LLaDA-8B-Instruct and Dream-7B-Instruct and report both sides of the trade: the strongest setting reaches 0.80 and 0.90 TPR at 1% FPR — on LLaDA at the best perplexity ratio of any method compared, on Dream above every token-local baseline under shared settings — whereas verification spends 144 masked evaluations and 10% re-denoising collapses TPR at $R = 1$ and halves it at the strongest setting (Section 4).

2 Related Work

Diffusion language models. Discrete and masked diffusion objectives carry diffusion [16] over to text [3, 29, 31, 36, 37] and now support instruction-following systems such as LLaDA and Dream [30, 41, 44], with commercial deployments following the same recipe [25]. Because the order in which positions commit affects quality and throughput [2, 6, 20, 38, 39], and because a run can equally be stopped before its step budget is spent [26], accelerators treat the whole schedule as a tunable degree of freedom. Shibboleth, by contrast, treats it as an invariant and draws on masked-token reconstruction only to define a probe it can later repeat.

Watermarking autoregressive text. Autoregressive watermarks bias keyed token subsets [21] or couple sampling to keyed randomness [1, 13, 24], with later work characterising their reliability and undetectability [10, 22, 43]. Detectors of this family are inexpensive precisely because a key–token hash needs no generator, yet that hash is defined over a resolved prefix and does not transfer unchanged to a context in which most token positions are still masked.

Watermarking diffusion language models. DLM-specific methods restore a well-defined keyed partition by other means, whether by removing prefix dependence [9], by averaging a red–green bias over unresolved context [14], by keyed sampling [4], or by exploiting the decoding order itself [17, 33, 40]. We compare against dgMARK, the strongest order-steering baseline with a zero-inference detector. Shibboleth differs from all of them in where the evidence resides, since dgMARK spends the DLM's freedom over commit order and reads the outcome off the text, whereas Shibboleth spends nothing from the schedule and instead has the verifier rerun the exact checkpoint, buying a statistic bound to model behaviour at the price of a detector that no longer runs on text alone.

Detector statistics. Detector design has moved past the asymptotic z-test: exact tails (binomial for green counts, Gamma for Gumbel sums) close the gap between nominal and empirical FPR that Gaussian approximations leave open at small α [12], permutation tests give non-asymptotic p-values for otherwise intractable alignment statistics [24], and windowed maxima must either pay an explicit union-bound correction or be calibrated empirically [22]. When edits leave watermark evidence in only a fraction of the text, sum-based pooling is provably dominated by statistics that concentrate on the surviving fraction [27]. Shibboleth's binomial tail belongs to the exact-test family, and its block structure admits the same sparse-signal statistics; Section 4.2 measures them and finds that under uniformly random edits the damage is dense and pooling wins, exactly the boundary the theory draws.

Detection that consults the model. Every DLM watermark above detects from the text and key alone, but detectors that pay more for their evidence are not new. Kuditipudi et al.'s edit-distance detector costs quadratic alignment under thousands of permutation resamples [24]; with model access, the Neyman–Pearson score strictly dominates the key-only Gumbel statistic [12], and log-likelihood-ratio detection reads both the watermarked and base conditionals [18]. GaussMark places its key in a Gaussian perturbation of the weights and verifies with one forward–backward pass of the generator [7], and watermarks built to expose model theft assume the verifier can query the suspect system [42]. Unlike GaussMark, which ships a perturbed checkpoint, Shibboleth leaves the released weights as they are and puts the key in the questions asked of them, and one deployed model therefore serves every document key.

3 Shibboleth: Watermarking by Model Response

3.1 Decoding and Notation

Masked diffusion language models differ from autoregressive models in that they predict a missing token from an arbitrary subset of revealed tokens, and the same sequence can consequently be produced under many decoding orders. Let x be a prompt and $y = (y_1, \ldots, y_N)$ the continuation to be generated, and for a revealed set $I \subset \{1, \ldots, N\}$ with a target position $i \notin I$, the model supplies a predictor $p_\theta(y_i \mid y_I, x)$ while the remaining positions carry the mask symbol, so that one sequence evaluation returns that distribution at every unresolved position at once.

Practical decoders nevertheless leave much of that freedom unused, and we study LLaDA's semi-autoregressive schedule, in which the continuation is partitioned into blocks that finish from left to right while diffusion resolves the masks inside the active block [2, 30], and the reference confidence rule decides which positions commit at each step. Everything below is therefore stated for one block B_b with the prefix F_b that precedes it.

3.2 Problem Setup

Our goal is a watermark whose detector consults the model rather than the text alone, telling a continuation generated under a secret key from one generated without it at a false-positive rate the verifier states in advance, while disturbing the decoding schedule and the output quality as little as the channel allows.

Embedder and verifier. The embedder receives the model, the prompt, a secret key K, and the target bits, whereas the verifier later receives only the completed sequence, the same model, K, and the public parameters, without logits, random seeds, or commit order. Each document then draws its own key from K and a fresh nonce through a keyed hash, $K_d = \mathrm{HMAC}(K, \mathrm{nonce}_d)$, which to anyone without K behaves as a random function of the nonce [5]. That document key generates both the probe family and the accepted answers, and the false-positive guarantee covers any text produced independently of it.

Threat model. Both the model and the algorithm may be public, since secrecy rests entirely in K. Concretely, the detector does not accuse innocent text, meaning that for every text written independently of the document key, machine text from another sampler included, a detector run at nominal level α fires with probability at most α, and the bound holds at finite sample size and under carriers the method itself selects. We claim nothing in the complementary direction, and an adversary willing to edit the output can therefore suppress detection, as Section 4.2 measures. We likewise claim nothing after key disclosure, under the detector-query access that recovers a key's behaviour [19], or when verification runs a numerically different checkpoint.

Post-hoc substitution. An embedder that swapped tokens in the finished text would be cheaper than one intervening during sampling, and we begin by measuring how much room such a swap has. Write τ for the likelihood budget of a substitution, the number of nats by which a replacement may fall below the live argmax. Across 36 settings that replace tokens after ordinary generation, the median selected position admits exactly one token at $\tau = 3$, the mean rising only from 1.6 at $\tau = 3$ to 5.6 at $\tau = 6$, and the best setting inside a $1.35\times$ perplexity budget reaches no more than 0.10 TPR at 1% FPR. Conditionals this sharp leave too little room for a forced replacement, and the embedder must therefore write while the conditional is still live.

We accordingly formalise the problem with three requirements.

1. **Replayability.** The verifier must be able to put exactly the question the embedder put, from the finished text and the key alone, with no generation trajectory, stored logits, or record of rejected samples.
2. **Blind selection.** The set of positions that carry a bit may depend arbitrarily on the text and the model, but that choice has to be settled before the sign the key rewards is consulted, since otherwise the reported false-positive rate understates the true one.

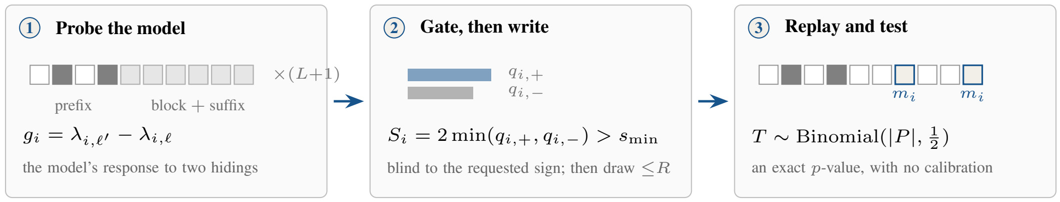

Figure 2: **How Shibboleth writes a mark and reads it back.** The same three steps run at generation and at verification, which is what makes the evidence replayable, and the gate in the middle step is settled before the key names the sign it accepts, which is what keeps its p-values exact.

3. **Schedule preservation.** The block partition and the confidence order must survive unchanged, so that a production decoder keeps the latency and quality profile it was tuned for.

The first two pull against each other, since replayability wants every scored context recoverable from a text the model helped choose while blind selection wants that choice to say nothing about the sign the key rewards. Sections 3.3 to 3.5 meet each requirement with one mechanism.

3.3 Probing the Model and Gating the Carriers

Unlike dgMARK, which obtains its channel by choosing which mask is resolved next, Shibboleth leaves the decoding schedule as the reference decoder set it. For block B_b, let F_b be the prompt together with the completed prefix, and let θ denote the reference checkpoint with p_θ for its predictor. Shibboleth then constructs L equal-size near, far, and keyed pseudorandom *probe patterns* $D_{b,\ell} \subset F_b$ indexed by $\ell \in \{1, \dots, L\}$, each the set of prefix positions it hides, and it masks $D_{b,\ell}$ together with the current block and the suffix, and one model evaluation therefore yields, for every token position $i \in B_b$ and every vocabulary item v, the log-probability

$$\lambda_{i,\ell}(v) = \log p_\theta(v \mid F_b \setminus D_{b,\ell}; B_b \text{ and suffix masked})\,. \tag{1}$$

Probing one block therefore costs $L+1$ sequence evaluations, one per pattern plus an unprobed baseline, and the verifier later repeats the same set.

Re-masking the block and the suffix removes every token unavailable when the probe was first applied, which is what buys replayability: the verifier restores the same state from the final text alone, and a transient generation-time response becomes a document statistic. Masking only the probed prefix would fail on both sides, since token positions committed later in the same block would leak into verification-time logits while the future suffix was invisible during embedding.

For an unordered pair of pattern indices (ℓ, ℓ') and a candidate token z, define the contrast $g_i^{\ell,\ell'}(z) = \lambda_{i,\ell'}(z) - \lambda_{i,\ell}(z)$. Let p_i^{base} denote the block-masked conditional, the model distribution when B_b and the suffix are masked but no probe pattern is applied. The mass on each sign and the two-sided score built from them are

$$q_{i,\pm}^{\ell,\ell'} = \sum_z p_i^{\text{base}}(z)\mathbf{1}[\pm g_i^{\ell,\ell'}(z) > 0], \qquad S_i^{\ell,\ell'} = 2\min(q_{i,+}^{\ell,\ell'}, q_{i,-}^{\ell,\ell'}). \tag{2}$$

Shibboleth selects, independently at each token position, the pair maximizing $S_i^{\ell,\ell'}$ and stores its contrast as g_i in the bank $\mathcal{R}_b$, after which the gate admits position i whenever $S_i > s_{\min}$, subject to an optional per-block cap. Because $S_i^{\ell,\ell'}$ is unchanged when ℓ and ℓ' are swapped, the gate neither observes nor favours the orientation the key will request, which is what buys blind selection, and both the pair selection and the gating recompute from the block-masked state. Best-pair selection is moreover what makes the gate productive, raising the measured mean score from 0.28 for a fixed near/far pair to 0.45, although the score stays a symmetric proxy computed on the block-masked conditional while generation runs from an evolving live one.

Once the gate has admitted a token position, the key supplies an orientation $\epsilon_i \in \{-1, +1\}$ from a PRF domain separate from pattern generation, pair selection, carrier assignment, and payload grouping. A target $w_i \in \{-1, +1\}$ then carries the desired provenance or message bit, so that for a candidate z the shared embedding-and-verification predicate becomes

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \tag{3}$$

Should $g_i(z)$ vanish exactly, a second domain-separated fair bit defines $m_i(z)$, and although such ties cannot be steered, the tie rule keeps the detector's p-values exact.

3.4 Writing the Requested Answer

Shibboleth inherits the reference confidence order together with the live conditional p_i, and the token positions that the gate rejected are sampled normally. At an admitted carrier define

$$A_i = \Big\{ z : w_i \epsilon_i g_i(z) > 0 \ \text{ and } \ p_i(z) \geq \kappa \max_v p_i(v) \Big\}. \tag{4}$$

It draws from p_i up to R times and commits the first member of A_i, and after R misses it commits one fresh unchecked draw instead. Because the row is never recomputed, those retries cost no model evaluation.

That procedure changes the sampling distribution, and by a stated amount. Let $a_i = p_i(A_i)$ be the acceptance mass and let $q_i = \sum_z p_i(z) m_i(z)$, where $m_i(z)$ applies Eq. (3) and the tie rule to candidate z, so that q_i is the probability that an unconditional draw scores as a detector hit irrespective of the plausibility floor. The committed-token distribution is

$$\big(1 - (1 - a_i)^R\big)\, p_i(\cdot \mid A_i) + (1 - a_i)^R p_i, \tag{5}$$

and its detector-hit probability is

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R (1 - q_i). \tag{6}$$

Whenever $\kappa = 0$ and the conditional places no mass on exact ties, $a_i = q_i$ and Eq. (6) collapses to $1 - (1 - q_i)^{R+1}$, so that a_i and q_i diverge only under a floor or a tie rule, where an unchecked fallback may score as a hit without belonging to A_i. The floor meanwhile bounds each guided proposal's model NLL to within $-\log \kappa$ of the live argmax but constrains neither the fallback nor semantic quality, which is why being model-proposed remains a support statement rather than a distortion-free claim.

3.5 Watermark Detection

Given a completed text y, the verifier re-masks each block and suffix, rebuilds the probe from the fixed prefix, reapplies the gate, and evaluates $m_i = m_i(y_i)$ at the admitted positions, so that rejected proposals and the generation trajectory are irrelevant. For provenance-only detection, which is all we evaluate here, the admitted carriers form a presence pool P with $w_i = +1$, whereas a payload implementation would keep the group for each payload bit separate, so that a balanced message cannot cancel positive against negative targets. The presence statistic and its one-sided p-value are

$$T = \sum_{i \in P} m_i, \qquad p_{\text{det}} = \Pr\big[\text{Binomial}(|P|, \tfrac{1}{2}) \geq T\big]. \tag{7}$$

Proposition 1 (Exact false-positive control)**.** *Assume the PRF domains above behave as independent random functions. Fix any text y written without the document key, and condition on the carrier set, selected pairs, contrast values, payload roles, and target bits. The orientation bits (and tie bits where needed) remain independent fair coins. Consequently the m_i are independent $\text{Bernoulli}(\frac{1}{2})$ variables and $T \sim \text{Binomial}(|P|, \frac{1}{2})$ exactly, so that p_{det} in Eq. (7) is an exact p-value and thresholding it at α holds the false-positive rate at or below α for every document length and carrier count.*

The conditioning carries the result, since carrier count, selected pair, and contrast magnitude may all depend arbitrarily on the text and the model, yet each is fixed before the independent orientation bit is consulted. Flipping ϵ_i therefore flips m_i whenever $g_i(y_i) \neq 0$, the tie rule supplies a fresh fair bit when it vanishes, and the carriers stay independent of one another throughout. In particular, neither an asymptotic approximation nor a calibration corpus enters anywhere, unlike a z-score whose reference distribution has to be estimated. The guarantee does nevertheless assume text written without the key, together with independence among the several uses of the key, which the domain-separated hash supplies.

Method	Setting	PPL ratio ↓	TPR at FPR ↑ 1%	0.1%	Sequence evaluations ↓ Generation	Detection
LLaDA-8B-Instruct						
KGW [21]	$\delta = 1$, 512 tok	1.22×	0.90	0.83	512	**0**
dgMARK [17]	$k = 1$, 512 tok	1.45×	0.83	0.80	512	**0**
dgMARK [17]	+3-beam, 512 tok	1.21×	**0.93**	**0.93**	∼2048	**0**
Shibboleth	$R = 1$, 512 tok	0.97×	0.43	0.20	656	144
Shibboleth	$R = 8, \kappa = 0.1$, 512 tok	0.87×	0.70	0.60	656	144
Shibboleth	$R = 16, \kappa = 0.05$, 512 tok	**0.74**×	0.80	0.60	656	144
Dream-7B-Instruct						
KGW [21]	$\delta = 1$, 512 tok	1.84×	0.57	0.37	512	**0**
dgMARK [17]	$k = 1$, 512 tok	1.75×	0.37	0.27	512	**0**
dgMARK [17]	+3-beam, 512 tok	**0.71**×	0.30	0.30	∼2048	**0**
Shibboleth	$R = 1$, 512 tok	1.34×	0.40	0.27	656	144
Shibboleth	$R = 8, \kappa = 0.1$, 512 tok	0.79×	0.70	0.67	656	144
Shibboleth	$R = 16, \kappa = 0.05$, 512 tok	1.09×	**0.90**	**0.90**	656	144

Table 1: **Watermark detectability.** Detection rate, perplexity against each method's own matched unwatermarked baseline, and compute in sequence evaluations. All rows are local reproductions on one checkpoint and prompt protocol at 512 tokens; each method keeps its own decoding regime, so PPL ratios are comparable within a method and indicative across methods. A PPL ratio below one is not evidence of better semantics. Shibboleth rows use $n = 30$ documents ($n = 20$ for $R = 16$) on both models; Dream PPL ratios rest on the 8–13 pairs per arm that survive the decoded-length rule. Here δ is KGW's green-list logit bias and k is dgMARK's candidate count. Every method uses one setting on both models, tuned on neither: the Dream rows carry the LLaDA-chosen hyperparameters, so cross-model rows measure transfer, not each method's per-model optimum.

4 Experiments

Datasets and setup. We evaluate `GSAI-ML/LLaDA-8B-Instruct` [30] and Dream-7B-Instruct [41] in fp16 on a single TITAN RTX; Dream emits the prediction for position i at raw-logit index $i-1$, and one shift in the model wrapper restores the shared convention for generation and replay alike. Detectability is measured on C4 `realnewslike` [35] prompts truncated to 300 characters as dgMARK's protocol prescribes. Every row decodes 512 tokens in 32-token blocks at one denoising step per token, each method under its own decoding regime: Shibboleth and dgMARK on the native confidence schedule with full-vocabulary multinomial sampling at temperature 0.8 (dgMARK with top-k=3), KGW strictly left-to-right as its green list requires. On top of the reference sampler, Shibboleth adds $L = 8$ probe patterns, threshold $s_{\min} = 0.5$, retry budget R, and an optional floor κ, and we test provenance detection ($w_i = +1$) rather than payload recovery.

Metrics and baselines. Detectability is TPR at analytic FPR thresholds from Eq. (7), and quality is GPT-2-large [34] perplexity against an unwatermarked baseline drawn from the same decoder, reported as $\exp(\mathrm{mean}(\mathrm{NLL}_{\mathrm{wm}} - \mathrm{NLL}_{\mathrm{ctrl}}))$. That ratio measures the distortion each method adds to its own unwatermarked baseline rather than absolute text quality, and comparisons of ratios across methods are therefore indicative rather than controlled. Compute counts sequence evaluations, and each of the $512/32 = 16$ blocks accordingly needs $L + 1 = 9$ of them, hence 144 for verification against $512 + 144 = 656$ for generation. We compare against KGW [21] under left-to-right decoding and dgMARK [17] under confidence decoding.

4.1 Main Results and Analysis

Retry budget and the floor. Detection scales with the number of guided draws as Eq. (6) predicts, since $R = 1$ reaches 0.43 TPR at 1% FPR and raising R from 2 to 4 to 8 on 30 held-out prompts lifts it from 0.73 to 0.83 to 0.87. Retries cost no model evaluation, and the budget is therefore paid in quality instead, with unfloored perplexity ratios climbing from $1.14\times$ at $R = 2$ to $1.40\times$ at $R = 4$ and $1.61\times$ at $R = 8$. The floor converts part of that back, since $(R, \kappa) = (8, 0.1)$ obtains 0.70 TPR at $0.87\times$ and $(16, 0.05)$ obtains 0.80 at $0.74\times$, both below their unwatermarked baselines, although a ratio below one indicates only that GPT-2-large prefers the high-probability tokens the floor admits.

Model	Method	Greedy decoding MMLU Acc ↑	Greedy decoding GSM8K Acc ↑	Greedy decoding HumanEval Pass@1 ↑	Multinomial decoding MMLU Acc ↑	Multinomial decoding GSM8K Acc ↑	Multinomial decoding HumanEval Pass@1 ↑
LLaDA-8B-Instruct	Non-watermarked	??	0.740	??	??	0.640	??
	KGW [21]	??	??	??	??	0.740	??
	dgMARK [17]	??	??	??	??	0.660	??
	dgMARK +3-beam	??	??	??	??	0.660	??
	Shibboleth	??	??	??	??	0.700	??
Dream-7B-Instruct	Non-watermarked	??	0.840	??	??	??	??
	KGW	??	??	??	??	0.700	??
	dgMARK	??	??	??	??	0.560	??
	dgMARK +3-beam	??	??	??	??	0.560	??
	Shibboleth	??	??	??	??	??	??

Table 2: **Text generation quality.** Accuracy on MMLU [15] and GSM8K [11] and pass@1 on HumanEval [8] under greedy and multinomial decoding, with the unwatermarked run of each model as the reference row of its block. Shibboleth uses $R = 16, \kappa = 0.05$; 50 problems per GSM8K cell, so differences within ± 0.1 are noise. Each watermark decodes under its own regime — KGW left-to-right (its green list requires a resolved prefix; its own control scores 0.640), dgMARK under confidence decoding (its argmax control is the greedy reference cell) — so rows are read against their own controls, not across. Cells marked **??** are pending measured runs.

Across models. The cross-model story inverts the LLaDA ranking. Carrying every method's LLaDA-chosen setting to Dream-7B-Instruct unchanged, both token-local channels weaken sharply: KGW falls to 0.57 TPR at 1% FPR at a $1.84\times$ perplexity ratio, since Dream under forced left-to-right decoding repeats itself heavily (84% duplicate bigrams), and dgMARK falls to 0.37 direct and 0.30 with three-beam lookahead. The model-response channel instead strengthens: the same $(16, 0.05)$ setting reaches 0.90 at both 1% and 0.1% FPR at a sync rate of 0.725 against 0.503 on the control, whose false positives stay at zero, and is on this model the strongest detector in the comparison. A token-local channel depends on how a particular decoder behaves under the constraint the watermark imposes, where a probe the model answers about its own output travels with the model. What does not transfer is the floor's perplexity dividend, since on Dream $(8, 0.1)$ reads $0.79\times$ but $(16, 0.05)$ reads $1.09\times$ — which (R, κ) is quality-optimal is model-dependent — and the Dream ratios rest on the 8–13 pairs per arm that survive the decoded-length rule, so they carry wide error bars. These are single-setting reproductions, not per-model tuning, for every method alike.

Downstream quality. Perplexity ratios are a proxy, so we additionally measure GSM8K [11] accuracy on 50 test problems at 256 tokens, comparing the watermarked $R = 16, \kappa = 0.05$ sampler against its matched unwatermarked control under the identical decoding schedule (Table 2). The watermarked arm scores 0.700 against the control's 0.640, and the paired view shows five problems flipping right and two flipping wrong, a tie within noise at this cohort size rather than an improvement; the sub-control perplexity ratio is therefore not purchased with degenerate text. Detection on these short outputs is weak (0.12 at 1% FPR) since 256 tokens admit few carriers, and the control arm sits at its null (0.02 at 1%). The peers behave the same way under their own controls (Table 2): no method moves GSM8K accuracy beyond noise at this cohort size, and short task outputs starve every detector, not only ours — direct dgMARK also reads 0.12 at 1% FPR, its three-beam variant 0.44, and KGW at $\delta = 1$ embeds almost nothing detectable at all (mean $z = +0.46$, zero TPR), since a short, heavily constrained answer leaves its bigram channel few free choices. A watermark's headline TPR, measured on 512-token continuations, does not transfer to 256-token task outputs for any method here. On Dream the task-quality cost separates the methods more sharply than any perplexity ratio did: KGW leaves accuracy untouched (0.700 against its own control's 0.700) while barely embedding (0.14 at 1%), whereas dgMARK drops it from 0.840 to 0.560 in both variants — twenty-eight against forty-two problems of fifty, well beyond noise — for 0.18 and 0.66 TPR respectively.

Cost against token-local detectors. Detection trails the token-based detectors at matched length: dgMARK's three-beam variant reaches 0.93 TPR at both 1% and 0.1% FPR against 0.80 and 0.60 for Shibboleth, its direct variant reaches 0.83 and 0.80, and KGW reaches 0.90 and 0.83. Shibboleth instead carries the only perplexity ratio in the comparison that reads below its own unwatermarked baseline, $0.74\times$ against $1.22\times$ for KGW, $1.45\times$ for direct dgMARK and $1.21\times$ for three-beam dgMARK; every baseline row costs perplexity, whereas the floor discards exactly the low-probability

Attack	Strength	TPR at FPR ↑ 1%	0.1%
Weakest setting: $R = 1$, 512 tokens ($n = 20$)			
None	—	0.35	??
Re-denoising (same model, argmax)	10% of positions	0.05	??
Deletion	10% of positions	0.05	??
Insertion	10% of positions	??	??
Substitution (random)	10% of positions	??	??
Copy–paste	25% kept, 1 segment	??	??
Paraphrase (DIPPER) [23]	lex 20, order 0	??	??
Strongest setting: $R = 16, \kappa = 0.05$, 1024 tokens ($n = 16$)			
None	—	0.88	0.88
Re-denoising (same model, argmax)	5% of positions	0.88	0.75
	10% of positions	0.50	0.38

Table 3: **Robustness to post-editing.** Detection after each attack under the deployed pooled detector, scored against the same analytic thresholds as Table 1. Evidence written into a replayable probe is synchronised to the prefix, so at $R = 1$ a 10% edit invalidates nearly every carrier; at the strongest setting the carrier redundancy absorbs a 5% edit outright and a 10% edit halves detection rather than erasing it. Small cohorts: one document is 0.05 ($n = 20$) or 0.06 ($n = 16$) of TPR.

draws a proxy scorer punishes. KGW additionally requires strict left-to-right decoding, under which this checkpoint repeats itself heavily, and its repeated bigram events must be deduplicated: without deduplication its control measured 37% FPR, and even with it two of thirty 512-token null documents cross the 0.1% threshold, a residual calibration failure of exactly the kind an exact conditional null rules out by construction [12]. Shibboleth alone, against all of that, spends 144 masked evaluations to detect at all, although paying for detection is not unprecedented: edit-distance detection runs a quadratic alignment under thousands of permutation resamples [24], and GaussMark verifies with a forward–backward pass of the generator itself [7]. Cohorts of twenty to fifty documents leave a standard error near 0.08 on a TPR of 0.80, and two rows separated by less than about a tenth should accordingly be read as a tie.

4.2 Calibration and Robustness

Calibration under the shipped rule. Proposition 1 is analytical, which is why we also measure the rule as deployed. Across 20 keys and 120 non-watermarked generations apiece, the mean FPR is 0.0121, 0.0037 and 0.0000 at nominal 0.10, 0.05 and 0.01, and the worst individual key gives 0.0583, 0.0250 and 0.0000, and control consequently holds for every key rather than on average alone, which is what binds when one key serves many documents.

Post-editing. Edit robustness is the channel's principal failure mode, and it degrades abruptly, since same-model argmax re-denoising of a random 10% of token positions takes TPR at 1% FPR from 0.35 to 0.05 on 20 $R = 1$ documents and deleting 10% measures the same. The collapse is one of synchronisation rather than signal strength, because an early edit alters the fixed prefix from which every later probe is rebuilt, and a single change thus invalidates every carrier after it. That cohort uses $R = 1$, and the headline setting at $R = 16$ consequently remains unattacked, which we record as an open gap rather than as evidence of resilience. Redundancy, however, changes the picture: attacking the strongest setting ($R = 16, \kappa = 0.05$ at 1024 tokens, $n = 16$) rather than the weakest, re-denoising 5% of positions leaves TPR at 1% FPR unchanged at 0.88 and re-denoising 10% halves it to 0.50 instead of erasing it, with false-positive control holding on the identically attacked controls. We also measured the block-local repair the pooled statistic seems to invite: a per-block exact test combined by Bonferroni over the blocks keeps the document-level guarantee by union bound, yet it never beats pooling here — 0.38 against 0.50 at 1% FPR under the 10% attack, and 0.69 against 0.88 clean — because uniformly random edits damage every block, so the sparse-signal regime in which concentration-type tests dominate sum-based ones [27] never arises. Bounding the conditioning window is the corresponding structural repair on the embedding side: with a 128-token window the response bank reads only the prompt and the last 128 generated tokens, so one edit can perturb at most five blocks' banks instead of every later one. Measured at $R = 16, \kappa = 0.05$ and 512 tokens on shared prompts and seeds ($n = 16$), the window costs nothing clean — 0.69 TPR at 1% FPR in both arms, at a comparable perplexity ratio — but buys nothing under random 10% re-denoising either (0.44 against 0.50, a one-document difference), and for the same reason: the window bounds propagation, while uniformly random edits already damage every window. Both structural repairs therefore stand or fall on the localized-edit test, which is the one still outstanding.

5 Conclusion

Shibboleth argues provenance from a keyed probe that a masked DLM answers about its own output rather than from a pattern left in the tokens, and because the gate is settled before the key names the sign it accepts, carriers may be discovered adaptively at no cost to the detector's exact p-values. That trade is worth making where the decoding schedule must not change and the evidence wanted is of the kind a specified model can vouch for, which a key–token hash cannot supply, since anyone holding the key can compute it without the generator. Absent either constraint, a token-local watermark is cheaper and easier to resynchronise after edits.

Limitations. That guarantee is nevertheless one-sided, since soundness holds for any text written without the key while nothing is claimed in the other direction, and editing degrades detection: a single same-model denoising pass over a tenth of the positions takes TPR from 0.35 to 0.05 at $R = 1$, and even at the strongest setting it halves detection (0.88 to 0.50 at 1% FPR, $n = 16$), so Shibboleth claims resilience to light editing at high redundancy and nothing beyond it. Cost is the second boundary, since 144 masked evaluations per document rule out screening a corpus. The evidence is bound to one checkpoint as well, and a verifier holding different weights, or the same weights at a different precision, consequently cannot reproduce the responses at all. Finally, the evaluation is itself narrow, covering two masked diffusion models, one prompt corpus, cohorts of twenty to fifty documents, quality proxies read as ratios against each method's own unwatermarked baseline, and provenance detection only, with the payload construction of Section 3.5 never measured.

Future work. Each boundary suggests its own repair. Anchoring probe patterns to content rather than to absolute positions would stop an edit from desynchronising every carrier after it, which is the single change most likely to turn an abrupt collapse into a graceful decline, and the remaining attack to run is the localized one — contiguous edits and copy–paste — where a per-block min-statistic should finally beat the pooled count. Verification cost is the easier target, since the binomial tail is monotone in the matches so far and a sequential test could stop once the evidence is decisive. Loosening the checkpoint requirement in turn asks for a contrast quantised coarsely enough that numerical drift cannot flip a sign, trading a little capacity for portability. Adaptive adversaries deserve their own study, since a detector answering queries leaks something about its key. Table 1 finally argues for composition rather than a winner, with a cheap token-local screen deciding what is worth a model-response check.

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